GREENS
TECHNOLOGIES

Course syllabus

DevOps Training

Learn DevOps from Linux basics to production. Master Git, GitHub Actions, Docker, Kubernetes, Terraform, AWS and monitoring on real-world projects, with classes in English and Tamil.

60+ hrs
Hands-on training

2
Live projects

English & Tamil
Classes available

Book a free demo class | +91 89399 15577 | contact@greenstechnologys.com

Greens Technology, Coimbatore, Tamil Nadu | Syllabus version: October 2026

Career paths

Choose the roadmap that matches your goal.

Every path starts with Linux, Shell and Git and ends with the capstone. The tracks in between change with the role you want.

Path 1 · Core

DevOps Engineer

You want a general DevOps role: automating builds, releases and infrastructure for any team or cloud.

Track 1	Linux, Shell and Git	8 hrs
Track 2	CI/CD with GitHub Actions and Jenkins	8 hrs
Track 3	Docker	6 hrs
Track 4	Kubernetes and Helm	10 hrs
Track 5	Terraform and Ansible	8 hrs
Track 7	Prometheus, Grafana and ELK	6 hrs
Track 9	End-to-end capstone	8 hrs

7 tracks · 54 hrs

Optional add-on: Track 8 DevSecOps (4 hrs).
Capstone focus: a complete pipeline from commit to a monitored Kubernetes deployment.
Certifications to target: Terraform Associate, CKAD (optional)
Typical roles: DevOps Engineer, Build and Release Engineer, Automation Engineer

Path 2 · Cloud

Cloud DevOps (AWS)

You want to build and run delivery pipelines and infrastructure on AWS.

Track 1	Linux, Shell and Git	8 hrs
Track 2	CI/CD with GitHub Actions and Jenkins	8 hrs
Track 3	Docker	6 hrs
Track 5	Terraform and Ansible	8 hrs
Track 6	AWS for DevOps	8 hrs
Track 7	Prometheus, Grafana and ELK	6 hrs
Track 9	End-to-end capstone	8 hrs

7 tracks · 52 hrs

Optional add-on: Track 8 DevSecOps (4 hrs).
Then add an AWS Associate track and work towards AWS DevOps Engineer – Professional .
Certifications to target: AWS Associate, AWS DevOps Engineer – Professional
Typical roles: Cloud DevOps Engineer, AWS DevOps Engineer, Cloud Engineer

Path 3 · Platform

Kubernetes and Platform

You want to run container platforms: clusters, Helm releases, observability and security.

Track 1	Linux, Shell and Git	8 hrs
Track 2	CI/CD with GitHub Actions and Jenkins	8 hrs
Track 3	Docker	6 hrs
Track 4	Kubernetes and Helm	10 hrs
Track 5	Terraform and Ansible	8 hrs
Track 7	Prometheus, Grafana and ELK	6 hrs
Track 8	DevSecOps	4 hrs
Track 9	End-to-end capstone	8 hrs

8 tracks · 58 hrs

Spend the most time on Track 4 and prepare for the Kubernetes exams. Capstone focus: autoscaling, monitoring and security scans on Kubernetes.

Certifications to target: CKA, CKAD, Terraform Associate

Typical roles: Platform Engineer, Kubernetes Administrator, Site Reliability Engineer

Tools you will work with

Linux · Git · GitHub Actions · Jenkins · Docker · Kubernetes · Helm · Terraform · Ansible · AWS · Prometheus · Grafana · ELK

Track details

Duration, prerequisites, projects and curriculum.

Track 1 · Foundations

Linux, Shell and Git

8 hrs
4 modules

Duration
8 hrs

Prerequisites
None. No prior IT experience needed.

Project
Write a shell script that automates a server health check and backup, and manage it in a GitHub repository using branches and pull requests.

Curriculum modules

01 **DevOps fundamentals**
DevOps culture, the delivery lifecycle, Agile, and how CI/CD fits into a team.

02 **Linux essentials**
File system, users and permissions, processes, package managers, systemd, SSH and basic networking (IP, DNS, ports, HTTP).

03 **Shell scripting**
Variables, conditions, loops, functions, error handling, cron jobs and automating routine admin tasks.

04 **Git and GitHub**
Commits, branching, merging, pull requests, code review and GitHub repository workflows.

Track 2 · Build and ship

CI/CD with GitHub Actions and Jenkins

8 hrs
5 modules

Duration
8 hrs

Prerequisites
Linux and Git basics (Track 1 or equivalent).

Project
Live project 1: a pipeline that builds, tests, scans and packages an application on every push, with approvals before release.

Curriculum modules

01 **CI/CD concepts and build tools**
Pipelines, artifacts, versioning and building Java or Node projects with Maven or npm.

02 **GitHub Actions**
Workflows, events, runners, secrets, matrix builds, caching and reusable workflows.

03 **Jenkins**
Installation, jobs, Jenkinsfile pipelines, agents, plugins and credentials.

04 **Quality gates**
Unit tests, code-quality analysis with SonarQube and failing builds on quality rules.

05 **Release strategies**
Environments, approvals, rolling and blue-green deployments, and rollbacks.

Track 3 · Build and ship

Containers with Docker

6 hrs
4 modules

Duration
6 hrs

Prerequisites
Linux and Git basics (Track 1 or equivalent).

Project
Containerize a multi-service application with Docker Compose and publish its images to a registry.

Curriculum modules

01 **Container fundamentals**
Images, containers, layers and how containers differ from virtual machines.

02 **Dockerfiles and image optimization**
Writing Dockerfiles, multi-stage builds, smaller and safer images.

03 **Networking and storage**
Bridge networks, port mapping, volumes and bind mounts.

04 **Docker Compose and registries**
Multi-container apps, environment configuration, tagging and pushing to a registry.

Track 4 · Run at scale

Kubernetes and Helm

10 hrs
5 modules

Duration
10 hrs

Prerequisites
Docker (Track 3 or equivalent).

Project
Deploy an application to a Kubernetes cluster with Helm, with ingress, autoscaling, health probes and rolling updates.

Curriculum modules

01 **Architecture and core objects**
Control plane, nodes, Pods, Deployments, ReplicaSets and Services.

02 **Configuration and storage**
ConfigMaps, Secrets, volumes, PersistentVolumes and claims.

03 **Networking and scaling**
Ingress, resource requests and limits, probes and the Horizontal Pod Autoscaler.

04 **Helm**
Charts, values files, releases, upgrades, rollbacks and chart repositories.

05 **Cluster operations and exam practice**
RBAC, upgrades, troubleshooting and timed practice for CKA and CKAD.

Track 5 · Run at scale

Infrastructure as Code: Terraform and Ansible

8 hrs
5 modules

Duration
8 hrs

Prerequisites
Linux basics and cloud basics (AWS Cloud Practitioner level).

Project
Provision a cloud network and servers with Terraform, then configure them with Ansible playbooks.

Curriculum modules

01 **IaC concepts**
Declarative infrastructure, version control for infrastructure and drift.

02 **Terraform fundamentals**
Providers, resources, variables, outputs, data sources and state.

03 **Terraform in practice**
Modules, remote state, workspaces and running Terraform from a CI pipeline.

04 **Ansible**
Inventories, playbooks, roles, handlers and idempotent configuration.

05 **Exam practice**
Practice questions and a mock test for the Terraform Associate certification.

Track 6 · Run at scale

AWS for DevOps

8 hrs
5 modules

Duration
8 hrs

Prerequisites
Linux and Git basics, and AWS Cloud Practitioner level knowledge.

Project
Deploy an application on AWS behind a load balancer using IAM roles, containers and an automated release pipeline.

Curriculum modules

01 **AWS core services for DevOps**
IAM, VPC, EC2, S3 and security groups.

02 **Containers and compute on AWS**
ECR, ECS, EKS and an introduction to Lambda.

03 **Delivery on AWS**
CodeBuild, CodePipeline, CodeDeploy and an introduction to CloudFormation.

04 **Operations and security**
CloudWatch, CloudTrail, Secrets Manager and cost awareness.

05 **Path to AWS DevOps Professional**
Exam domains overview and how to prepare after an AWS Associate certification.

Track 7 · Operate and secure

Observability: Prometheus, Grafana and ELK

6 hrs
5 modules

Duration
6 hrs

Prerequisites
Linux basics, plus Docker or Kubernetes basics.

Project
Build dashboards and alerts for a containerized application with Prometheus, Grafana and Alertmanager, and centralize its logs with the ELK stack.

Curriculum modules

01 **Monitoring fundamentals**
Metrics, logs and traces, SLIs, SLOs and alerting principles.

02 **Prometheus**
Exporters, scraping, labels and PromQL queries.

03 **Grafana and alerting**
Dashboards, Alertmanager rules and notification channels.

04 **Centralized logging**
Elasticsearch, Logstash, Kibana and log shipping with Filebeat.

05 **Monitoring Kubernetes**
Cluster and workload metrics, and dashboards for Pods and nodes.

Track 8 · Operate and secure

DevSecOps

4 hrs
4 modules

Duration
4 hrs

Prerequisites
CI/CD and Docker basics (Tracks 2 and 3 or equivalent).

Project
Add secret scanning, dependency checks, container image scans and policy gates to a CI pipeline.

Curriculum modules

01 **Shift-left security**
Where security fits in the pipeline and the common risks in code and containers.

02 **Code and dependency scanning**
Static analysis with SonarQube and dependency alerts.

03 **Container and IaC scanning**
Image scanning with Trivy and checking Terraform for misconfigurations.

04 **Secrets and policy**
Managing secrets safely and enforcing pipeline policies.

Track 9 · Capstone

End-to-end capstone project

8 hrs
5 modules

Duration
8 hrs

Prerequisites
Tracks 1 to 7 completed (Track 8 recommended).

Project
Live project 2: take an application from a GitHub commit to production. GitHub Actions builds and scans the image, Terraform provisions AWS infrastructure, Helm deploys to Kubernetes, and Prometheus and Grafana monitor it.

Curriculum modules

01 **Plan and architecture**
Choose the application, design the pipeline and agree the environments.

02 **Build the pipeline**
GitHub Actions workflows, tests, security scans and container images.

03 **Provision and deploy**
Terraform on AWS, Kubernetes with Helm, and release strategy.

04 **Observe and present**
Dashboards, alerts, a runbook, a demo and a review.

05 **Interview preparation**
Presenting your project, resume review and mock interviews.

How you can learn

Online, classroom at our Coimbatore centre, one-to-one and corporate training, with weekend batches.

Want batch timings and fees?

Call or message us. An advisor will share the latest batches, free demo class dates and fees, and help you pick the right path.

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Syllabus version: October 2026. Topics, duration and projects are as published on our website.